Aneesh Ganti

510-518-7807 | aneeshg5@illinois.edu | aneeshganti.dev | linkedin.com/in/aneesh-ganti | github.com/aneeshg5

EDUCATION

University of Illinois Urbana-Champaign Expected: May 2028
Bachelor of Science in Computer Science and Mathematics GPA: 3.75/4.00

Relevant Coursework: • Data Structures • Algorithms • Distributed Systems • Database Systems • Cloud Computing • Software Design • Machine Perception • Computer Vision • Machine Learning Systems • System Programming • Computer Architecture • Data Science

PROFESSIONAL EXPERIENCE

Software Engineer Intern | Brunswick Corporation January 2026 — Present
- Engineered regression and classification models for a propeller recommendation tool to save **$65,000+** in testing for **400+** boat users.
- Built progressive web app with TypeScript and Azure for test engineers to capture **5x data** and cut analysis runtime from **3 hrs** to **9s**.
- Evaluated frontier agentic AI tools on decoding **70+** encrypted CAN signals for cryptanalysis on helm, throttle, and fuel vulnerabilities.
- Patented wipeout detector for tow sports with YOLOv11 and tether kinematics to track riders at **97%** accuracy without human spotters.

Founding Engineer | PromoPigeon LLC (iVenture 12) February 2026 — Present
- Launched Gemini-powered agent to generate product mockups from client emails at scale and cut response time from **2 days** to **<60s**.
- Designed spaCy and PyMuPDF document parser to extract and store purchase order data and image assets for **2,500+ daily proofs**.
- Architected Dockerized artwork engine on AWS ECR as an API agent tool for mockup agent to eliminate manual graphic design work.

Software Engineer Intern | Tekweld Manufacturing May 2025 — February 2026
- Built sales outreach tool with Python and AWS to create and ship personalized samples to **700+** businesses with **14%** conversion rate.
- Accelerated product design from **1 wk** to **15 min** using serverless, LLM-driven pipeline for label printing and visual quality assurance.
- Shipped React and FastAPI web application on AWS Lightsail to manage order fulfillment and brand assets for **90,000+** client leads.

RESEARCH EXPERIENCE

CV & Robotics Researcher | University of Illinois Urbana-Champaign February 2026 — Present
- Developed a self-supervised framework with Meta's SAM and DINOv3 to train annotation-free segmentation in data-sparce domains.
- Designed ViT-conditioned diffusion model with PyTorch to forecast object trajectories from RGB images for **17s** robot motion planning.
- Integrated 2D-to-3D lifting with TAPIP3D keypoints to generate motion training labels from videos and scale training data by **260%**.

Machine Learning Researcher | University of California, Santa Barbara June 2023 — August 2023
- Trained TensorFlow and Keras DNN to predict battery degradation state and remaining useful life from **10,000+** NASA charge cycles.
- Reduced battery health prediction error by **18%** over BLS-RVM and LSTM baselines to validate generalization across battery models.
- Collaborated with UCSB and MIT AI researchers on EV battery diagnostics methods and presented findings at the capstone seminar.

PROJECTS AND ACTIVITIES

Software Engineer | Spaceshot Rocketry Team August 2024 — Present
- Implemented C++ Extended Kalman Filter to estimate rocket dynamics with **10**-ft accuracy in **75,000**-ft launch for flight tracking.
- Built telemetry interface with JavaScript to track MQTT transmissions and ensure zero packet loss in turret control during launches.

Software Contributor | ZenGM Sports Management January 2026 — Present
- Delivered **20+ features** using browser-native multi-threading and IndexedDB for a sports franchise simulation with **50,000+ users**.
- Developed real-world schedule generator with TypeScript to detect **4-type** match conflicts and balance league structure for any session.

Data Engineer | PhinD Experts (iVenture 10) May 2025 — August 2025
- Automated CI/CD with GitHub Actions, Docker, and AWS to deploy a PhD recruiter-candidate matching platform with **4,000+** profiles.
- Revamped Airflow ETL pipeline to scrape LinkedIn and Google Scholar and reduce PhD candidate matching time from **1 min** to **25s**.

Software & Simulations Lead | NASA Human Lander Challenge August 2024 — July 2025
- Engineered ML pipeline using Python, ANSYS, and Scikit-learn to improve in-space cryogenic propellant transfer efficiency by **84%**.
- Designed agnostic protocol for **NASA's Artemis program** and led team to win Best Presentation and **$1,200** prize over **30+** universities.

SKILLS

Programming Languages: Python, **C++**, **Java**, **SQL**, TypeScript, JavaScript, GoLang, HTML, CSS, R, Verilog, MIPS
Full-Stack & Backend: FastAPI, **PostgreSQL**, **REST APIs**, React, Node.js, Flask, MongoDB, Express, Redis, WebSockets
GenAI Tools: Hugging Face, **LangChain**, **LlamaIndex**, RAG, Pinecone, LangGraph, vLLM, PEFT, LoRA, ONNX, Claude Code
ML & Data Science: PyTorch, **Scikit-learn**, **TensorFlow**, CUDA, NumPy, Pandas, OpenCV, spaCy, SciPy, Matplotlib, MLflow
Cloud & Infrastructure: AWS, **Docker**, **Kubernetes**, CI/CD, Git, GitHub, GitHub Actions, Google Cloud, Linux, Jest, Agile